



EASWARI ENGINEERING COLLEGE



Smart Zonal Energy Optimization System for Educational Institutions

THE REAL PROBLEM

- ✓ In most Indian classrooms:
 - Fans and lights operate at full capacity
 - No control based on real occupancy
 - Manual switching leads to negligence
 - Energy waste accumulates daily
- ✓ Energy cost is a recurring institutional expense with no optimization mechanism.



PROBLEM SIZE (MARKET OPPORTUNITY)

Target Segment:

- Engineering colleges
- Private institutions
- Coaching centers

Total Colleges in India: 40,000

Average Classrooms per College: 20

Total Target Classrooms: 800,000

Product Price per Classroom: ₹8,000

Total Addressable Market (TAM):

₹6,400 crore

Even 1% Market Capture:

8,000 classrooms → ₹6.4 crore revenue potential



OUR SOLUTION



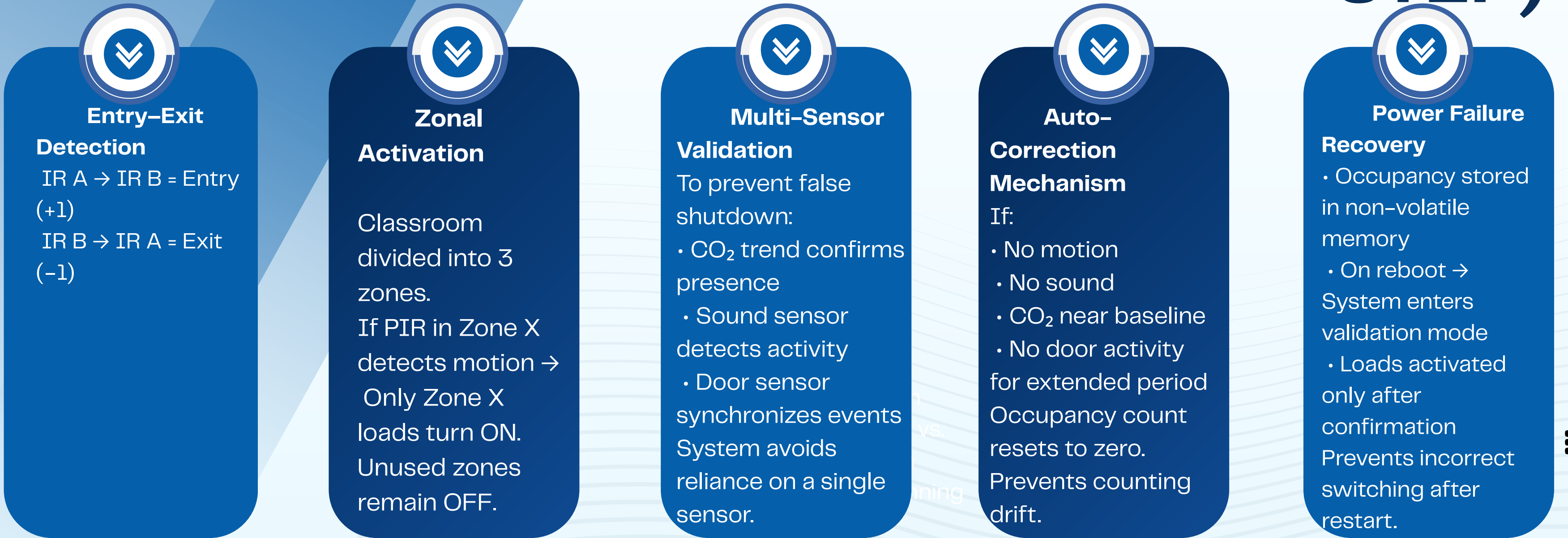
- ✓ We developed a:
Multi-Sensor Zonal Energy Optimization System that:
 - Detects real-time occupancy
 - Activates only required classroom zones
 - Validates presence using multiple
- ✓ sensors
 - Auto-corrects counting drift
 - Recovers safely after power failure
 - Works without cameras
- ✓ Designed specifically for Indian educational institutions.

TECHNOLOGY ARCHITECTURE

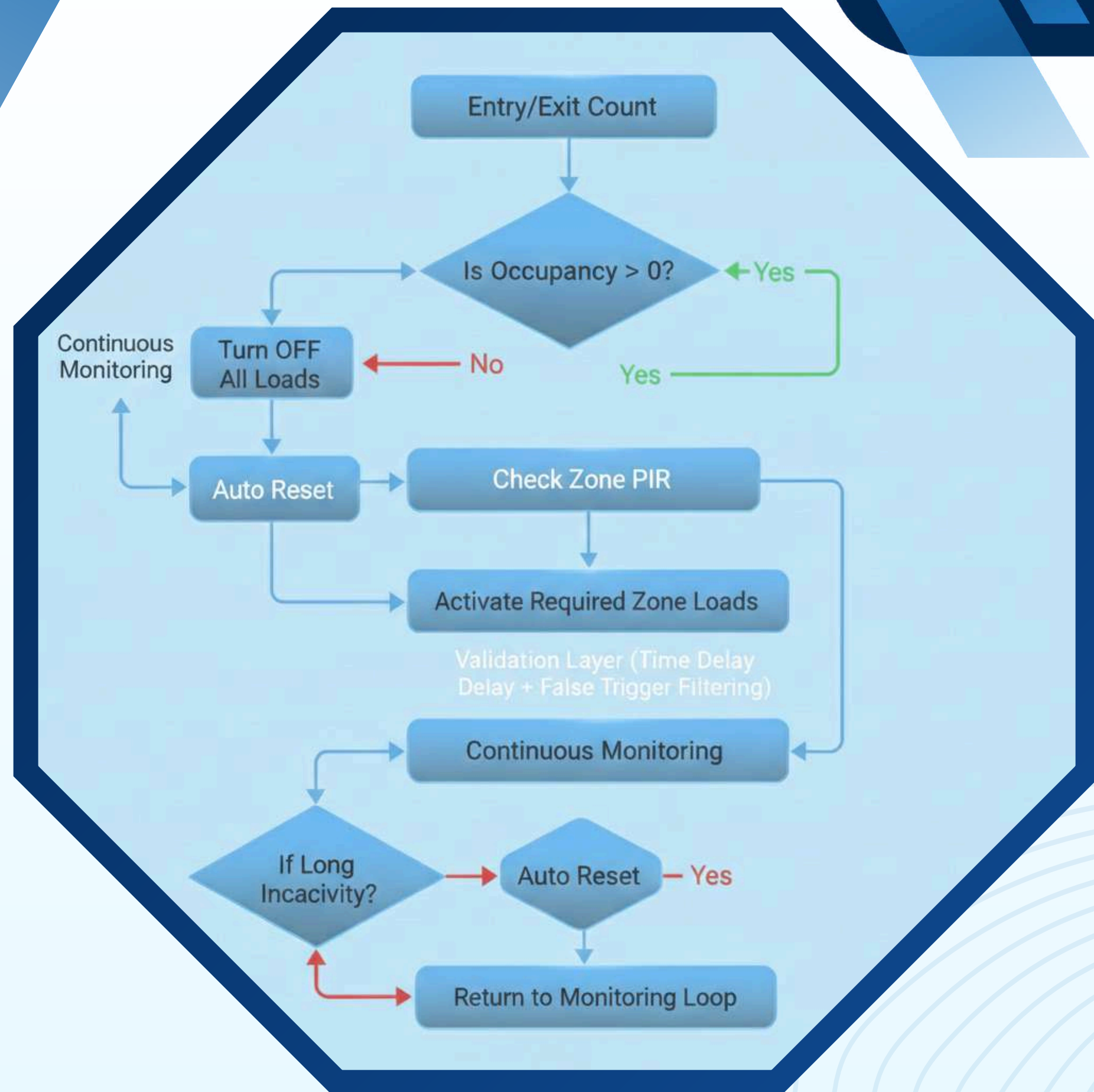
- ✓ **Controller:**
 - ESP32
- ✓ **Sensors:**
 - Dual IR Beam – Entry/Exit Counting
 - PIR Sensors – Zonal Motion Detection
 - CO₂ Sensor – Occupancy Validation
 - Sound Sensor – Low-Motion Detection
 - Door Reed Switch – Event Synchronization
- ✓ **Outputs:**
 - Relay modules controlling fan and lighting circuits
- ✓ **No camera. No privacy concerns.**

■■■■

HOW THE SYSTEM WORKS (STEP-BY-STEP)



Flow Chart



WHAT MAKES US DIFFERENT

- ✓ Most low-cost systems:
 - Use only PIR
 - Have no correction logic
 - Fail during low motion
 - ✓ Our System Uses:
 - Layer 1 – Directional Counting
 - Layer 2 – Zonal Detection
 - Layer 3 – Environmental Validation
 - ✓ Result:
 - Reduced false shutdown
 - Improved reliability
 - Higher operational accuracy
- We engineered reliability — not just automation.



FINANCIAL ANALYSIS



Typical Classroom Load:
770W average



Unnecessary operation (2
hrs/day):

≈ 1.54 kWh/day

≈ 462 kWh/year



At ₹8 per unit:

₹3,696 saved per classroom
annually



For 20 classrooms:

₹73,920 annual savings

Savings increase with zonal
control.

PRODUCT COSTING & PRICING



- ✓ Estimated Manufacturing Cost (bulk):
₹4,500 per unit
- ✓ Selling Price:
₹8,000 per classroom
- ✓ Gross Margin:
≈ ₹3,500 per unit
- ✓ Payback Period:
~2 years
- ✓ After payback → Direct savings for institution.



BUSINESS MODEL

Revenue Streams

- Hardware sales
- Installation charges
- AMC
- Future dashboard subscription

Go-To-Market Strategy

- Pilot in home institution
- Direct outreach to private colleges
- Energy audit demo model
- Electrical contractor partnerships

Trust Building

- No structural wiring modification
- Privacy-friendly design
- Documented ROI case study
- Faculty mentorship support

Customer Acquisition Cost

- Estimated ₹2,000–₹3,000 per institution (early stage)



TARGET CUSTOMERS



- ✓ Phase 1:
 - Private schools
 - Engineering colleges
 - Coaching centers
- ✓ Phase 2:
 - Government institutions
 - Smart campus projects
 - CSR-funded education initiatives



SCALABILITY STRATEGY

- ✓ Step 1 – Pilot in One Institution
- ✓ Step 2 – Measure and Document Savings
- ✓ Step 3 – Expand Within Campus
- ✓ Step 4 – Regional Deployment Model
- ✓ Modular design supports multi-classroom integration.



Risk Handling & Failure Management

Scenario	System response
Silent exam	Time based persistence
Mass exit	Drift Correction Logic
Power failure	Recovery mode
Sensor failure	Fall back to count logic
Window ventilation	Ignore Co2 as a primary



- Reduces energy wastage
 - Lowers carbon footprint
 - Promotes sustainable campus culture
 - Encourages responsible infrastructure usage
- Supports national energy conservation goals.

FUTURE ROADMAP

SOCIAL & ENVIRONMENTAL IMPACT



- Mobile monitoring application
- Real-time energy usage analytics
- Cloud-based dashboard
- Integration with solar systems

Long-term Vision:

Smart Campus Energy
Ecosystem.

Prototype Phase 1

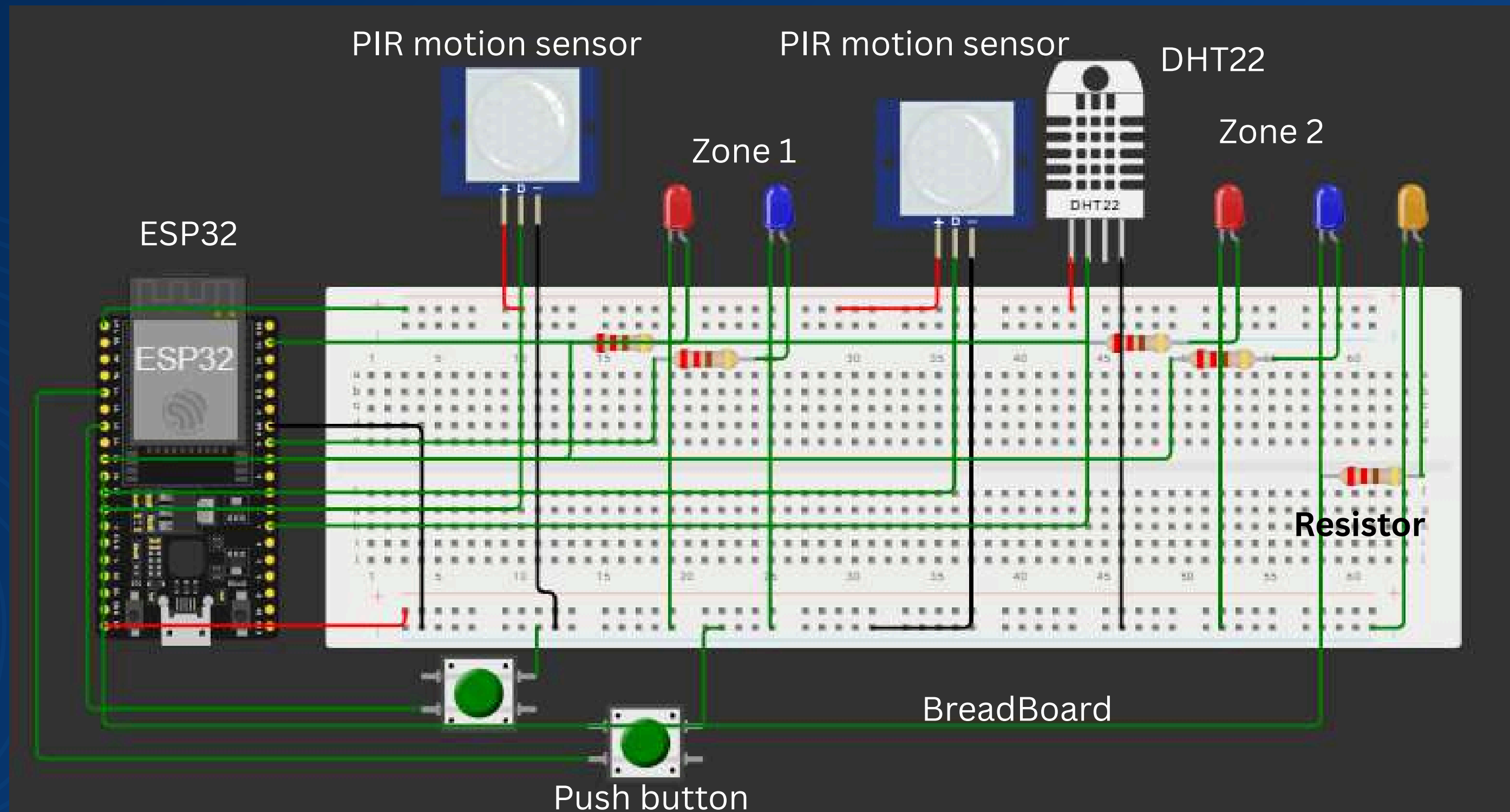
Prototype Overview:

The Phase 1 prototype demonstrates a Smart Zonal Energy Optimization System for classrooms using occupancy detection and automated device control. The system activates lights, fans, and ventilation only when the classroom is occupied, helping reduce unnecessary energy consumption.

Components:

- 2 PIR sensors
- 2 Red colour LEDs (lights)
- 2 Blue colour LEDs (fans)
- 1 Yellow Colour LEDs (Ventilation indicator)
- ESP32 Microcontroller
- 2 Push Button (Entry/Exit)
- DHT22 (temperature sensor)
- BreadBoard
- 220 Ohm Resistors

Prototype Image



ESP32 SNIPPET CODE

1. Entry–Exit Occupancy Counting

```
// Entry button pressed → increment student count
if (entryState == LOW && prevEntryState == HIGH) {
    students++;
    Serial.println("Entry detected!");
}

// Exit button pressed → decrement student count
if (exitState == LOW && prevExitState == HIGH) {
    if (students > 0) {
        students--;
        Serial.println("Exit detected!");
        if (students == 0) {
            zone1Active = false;
            zone2Active = false;
            ventilationActive = false;
        }
    }
}
```

2. Zonal Motion Detection Logic

```
// Read sensors
int motion1 = digitalRead(PIR1);
int motion2 = digitalRead(PIR2);
temperature = dht.readTemperature();

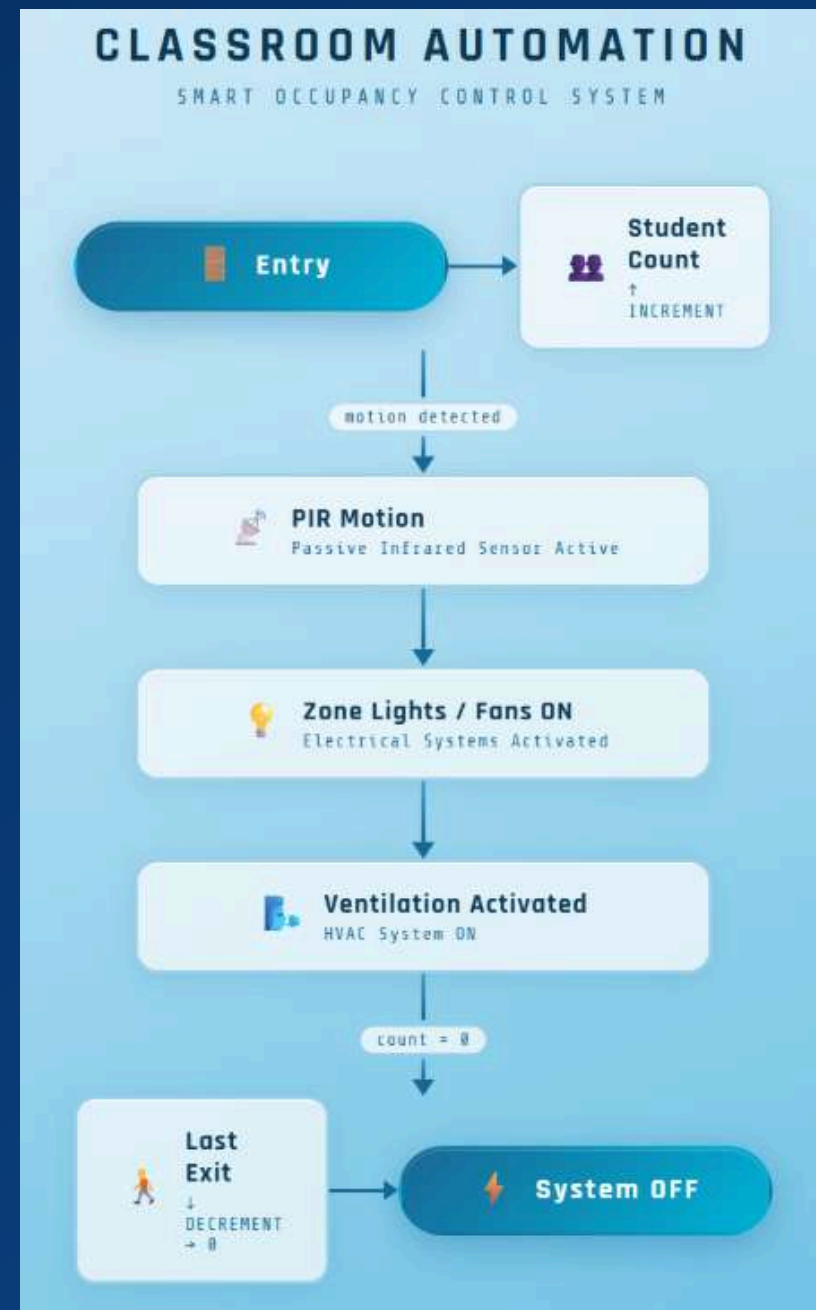
// Zone automation
if (students > 0) {
    if (motion1 == HIGH) zone1Active = true;
    if (motion2 == HIGH) zone2Active = true;
    if (motion1 == HIGH || motion2 == HIGH) ventilationActive = true;
} else {
    zone1Active = false;
    zone2Active = false;
    ventilationActive = false;
}
```

3. Smart Ventilation Control

```
if (students > 0) {
    if (motion1 == HIGH || motion2 == HIGH) ventilationActive = true;
} else {
    ventilationActive = false;
}

digitalWrite(VENTILATION_LED, ventilationActive ? HIGH : LOW);
```


Flow Chart



Working Principle

Key Functions Demonstrated

Occupancy Counting

- Entry and exit push buttons update the number of students in the classroom.

Zonal Automation

- PIR sensors detect motion and activate lights and fans only in occupied zones.

Smart Ventilation

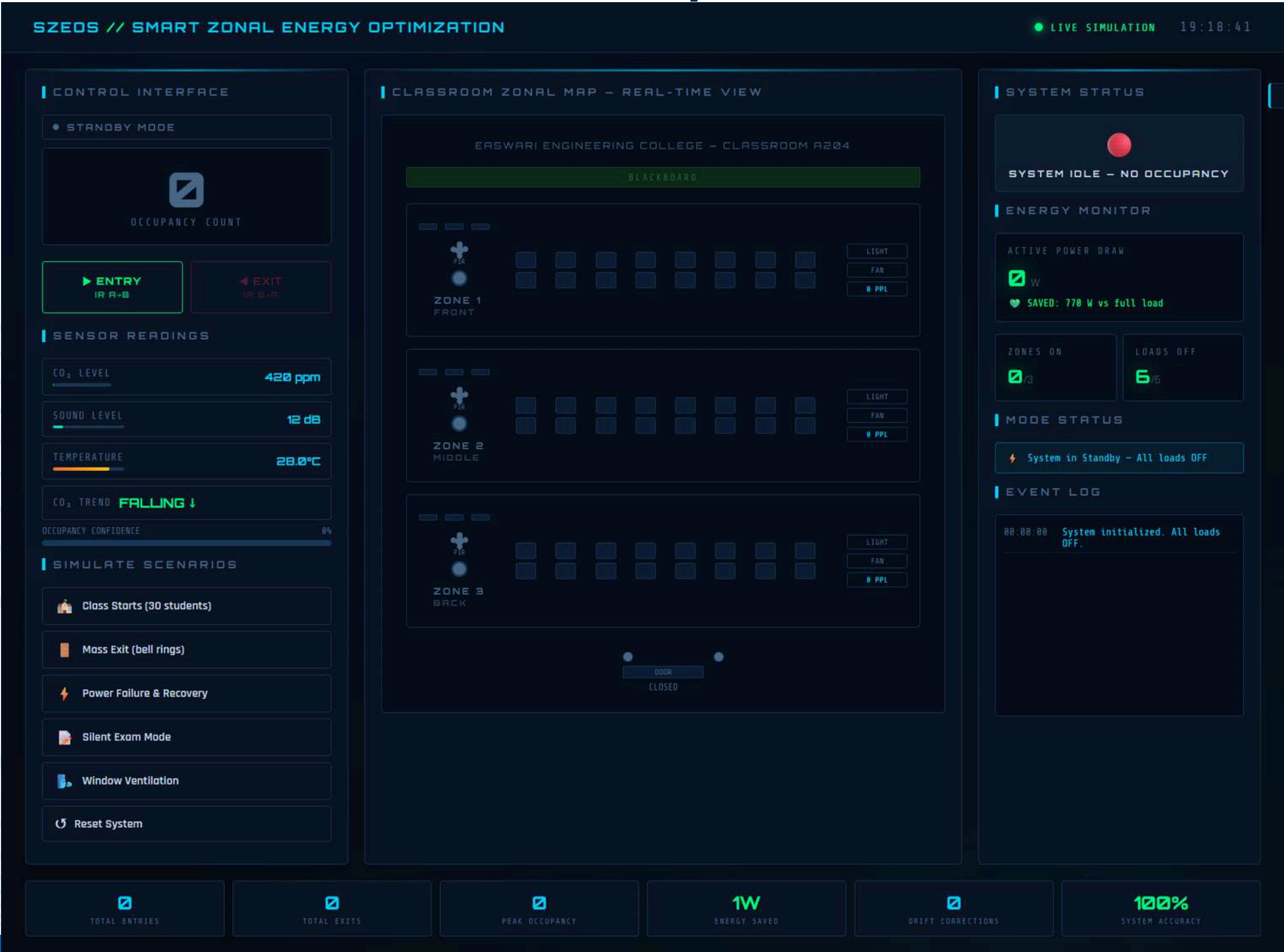
- Ventilation activates when motion is detected and remains active while the classroom is occupied.

Automatic Energy Saving

- When the last student exits, all lights, fans, and ventilation systems automatically turn OFF.



Smart Zonal Energy Optimization – Live class demo (Future Implementation)



Deployment Roadmap

Phase 1 – Functional Prototype (Completed)

- Zonal activation logic validated
- Entry-exit occupancy tracking

Phase 2 – Sensor Expansion

- CO₂ integration
- Sound-based validation
- EEPROM recovery logic

Phase 3 – Institutional Deployment

- Multi-classroom integration
- Central monitoring dashboard

We are presenting a validated prototype with a clear scalability pathway.





WHY THIS WILL SUCCEED

- ✓ Real institutional problem
- ✓ Measurable financial benefit
- ✓ Affordable pricing model
- ✓ Clear ROI
- ✓ Scalable deployment
- ✓ Technically feasible

We are not presenting a concept.

We are presenting a deployable product with market potential.

CONCLUSION

Our project presents a low-cost, practical solution for classroom energy optimization using basic sensors and zonal control.

The prototype validates that intelligent load management can reduce energy waste, improve efficiency, and provide a scalable model for educational institutions.

“We are not just reducing energy consumption — we are redesigning how classrooms use energy intelligently.”